

COIMBRA, Portugal

WMO: 085490

Lat: 40.2135N

Lon: 8.4552W

Elev: 35

StdP: 100.91

Time Zone: 0.00 (EUW)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.3	3.5	-3.1	2.9	7.3	-1.6	3.3	7.2	12.0	11.8	10.4	12.4	1.7	180	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	34.1	21.2	31.5	20.6	29.4	20.0	22.4	31.2	21.5	29.6	20.6	28.0	2.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.2	25.6	18.7	13.6	24.1	18.1	13.1	23.1	65.6	31.3	62.2	29.6	59.4	28.2	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	7.5	6.3	DB 0.3	38.7	1.2	1.3	-0.6	39.6	-1.2	40.4	-1.9	41.2	-2.7	42.1		
(5)			WB -1.1	24.7	1.3	1.6	-2.0	25.9	-2.7	26.8	-3.4	27.7	-4.4	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	10.0	11.1	13.5	13.6	17.2	19.6	22.4	22.1	20.4	16.9	13.6	11.6		
	DBStd	5.07	2.28	2.62	2.82	3.16	3.15	2.85	3.05	2.67	2.99	3.01	2.95	2.57		
	HDD10.0	73	27	17	5	4	0	0	0	0	0	0	6	14		
	HDD18.3	1269	257	202	152	146	59	17	2	2	13	66	143	209		
	CDD10.0	2278	29	48	112	112	224	289	386	376	312	214	114	63		
	CDD18.3	432	0	0	2	4	25	56	129	119	74	21	2	0		
	CDH23.3	3246	0	0	12	48	219	423	1102	873	472	98	1	0		
(13)	CDH26.7	1197	0	0	1	8	67	138	484	321	159	18	0	0		
(14)	Wind	WSAvg	2.3	2.5	2.4	2.5	2.3	2.2	2.1	2.0	2.0	2.4	2.4	2.8		
(15) Precipitation	PrecAvg	923	133	102	79	83	75	30	10	14	52	118	119	115		
	PrecMax	1269	387	291	290	238	187	92	50	53	136	268	416	358		
	PrecMin	549	10	16	0	28	0	1	0	0	1	19	9	8		
	PrecStd	210	93	69	68	50	48	22	12	14	37	66	86	78		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	20.7	25.0	27.8	31.7	33.4	37.0	36.6	34.9	29.1	22.4	18.6		
		MCWB	12.3	13.2	15.7	16.9	18.3	20.8	22.1	21.9	21.1	18.6	16.6	14.9		
	2%	DB	15.8	18.0	22.8	24.4	28.6	30.2	34.6	32.8	30.9	26.0	20.9	17.1		
		MCWB	12.2	12.1	15.0	15.6	17.9	20.5	21.7	21.0	20.1	17.5	16.0	14.2		
	5%	DB	14.7	16.5	20.6	21.6	25.6	27.9	31.8	30.1	28.0	23.8	19.3	16.2		
		MCWB	11.9	11.9	14.0	14.7	16.9	19.6	21.2	20.4	18.9	17.0	15.4	14.1		
	10%	DB	13.8	15.2	18.7	19.4	23.1	25.5	29.0	27.7	25.8	21.9	18.1	15.5		
		MCWB	11.4	11.7	13.0	13.6	16.3	18.5	20.5	19.8	18.3	16.4	14.7	13.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.4	17.1	18.0	19.7	22.8	23.9	23.0	21.9	19.5	18.6	16.5		
		MCDB	15.3	17.2	22.7	25.6	27.0	30.4	34.7	32.7	31.0	25.5	20.1	17.0		
	2%	WB	13.7	14.2	15.5	16.4	18.6	21.2	22.6	22.0	20.8	18.6	17.1	15.7		
		MCDB	14.6	15.9	21.3	22.7	26.4	28.9	31.7	30.4	27.7	23.4	19.1	16.4		
	5%	WB	12.9	13.2	14.5	15.2	17.6	20.1	21.6	21.1	20.0	17.8	16.0	14.8		
		MCDB	13.9	14.8	19.6	20.1	24.5	26.5	30.1	28.9	26.1	21.5	18.5	15.9		
	10%	WB	12.0	12.3	13.6	14.1	16.6	19.0	20.7	20.3	19.3	17.1	15.3	13.7		
		MCDB	13.2	14.2	17.6	18.2	22.2	24.6	28.5	27.0	24.4	20.3	17.5	14.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.5	9.4	9.2	9.9	10.5	12.1	11.8	10.7	7.9	7.1	5.8		
		MCDBR	8.0	9.5	12.8	13.2	14.3	14.6	16.8	15.7	14.9	11.4	8.5	6.2		
		MCWBR	5.2	5.3	6.7	6.4	6.7	6.7	6.8	6.4	6.3	5.1	4.8	4.2		
	5% WB	MCDBR	5.3	6.7	11.3	12.0	12.5	13.1	14.9	13.6	12.1	9.3	6.3	4.2		
		MCWBR	4.2	4.6	6.2	6.1	6.2	6.4	6.7	5.9	5.3	4.6	4.1	3.5		
(40) Clear-Sky Solar Irradiance	taub		0.330	0.343	0.377	0.377	0.384	0.389	0.377	0.394	0.382	0.369	0.340	0.327		
	taud		2.503	2.457	2.367	2.378	2.384	2.389	2.405	2.359	2.417	2.471	2.506	2.510		
	Ebn at Noon		833	872	874	897	894	887	894	869	858	831	813	805		
	Edh at Noon		76	91	111	117	119	119	116	118	105	89	76	71		
(44) All-Sky Solar Radiation	RadAvg		2.02	3.02	4.07	5.27	6.45	7.09	7.32	6.60	5.12	3.36	2.25	1.83		
	RadStd		0.32	0.41	0.55	0.62	0.60	0.45	0.31	0.28	0.40	0.36	0.37	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page